OPEN ITEMS CHECKLIST — part 6 (17 October 2026)

Changes since part 5 (16 October 2026). Stages 6a to 6c complete the Double-Number and Core Extension word sets and the `ENVIRONMENT?` query table. Every stage was assembled in all four builds (`SERIALPOLL` 0/1 by `UNITTESTS` 0/1) and with every test selector from 0 to 18.

Double-Number word set (stage 6a)

- ☑ **Added `D0=` , `D0<` , `D2*` , `D2/` , `M*/` , `2LITERAL` , `2VALUE`** , and `TO` now recognizes a `2VALUE` (it checks the behaviour cell at xt+3 for `DO2VALUE` ; a 2VALUE stores x2 at the lower address like `2CONSTANT`).
- ☑ **`M*/` method.** Take the magnitudes; form `|d1.lo|*|n1|` and `|d1.hi|*|n1|` with `UMUL32` and add them into a 3-cell product on an S frame; divide by `|n2|` in three 32/16 `UDIV32` steps (each quotient is under 65536); negate the result if an odd number of `d1` , `n1` , `n2` are negative. Symmetric division (truncates toward zero), `THROW -10` for a zero divisor. The method was checked against exact integers in a Python model (200000 random cases including the extremes) before it was written in assembler. Shadow entry `M*/W.1` .
- ☑ **Assembler unit tests** for `D0=` , `D0<` , `D2*` , `D2/` and `M*/` (including the divide-by-zero case) in the DArithmetic group (selector 4).
- ☑ **ANS test file `28_double_word.tests.fs`** built from the forth-standard.org Double-Number tests. Constants come from `D2/` , `M*/` and `MAX-D` instead of hard-coded values; typos fixed; the previously disabled blocks enabled. All tests pass in MAME. The `D.` / `D.R` test is visual (lines in pairs).

ENVIRONMENT? (stages 6a and 6b)

- ☑ **`CORE` , `CORE-EXT` , `DOUBLE` and `DOUBLE-EXT` answer.** A word-set query returns a value flag and then the recognized flag (`true true`). This matches the standard's table, where those queries return a flag, and gforth. An unknown name still returns a single `false` .
- ☑ **Section 24 tests cover the whole standard query table**, with `/HOLD` and `/PAD` checked against the standard minimums (34 and 84).

- ☑ The first version of the DOUBLE tests expected a single flag; the test was wrong, not the code.

Core Extension word set (stages 6b and 6c)

- ☑ **Added `.(`, `C"`, `SAVE-INPUT`, `RESTORE-INPUT` (6b), `S\"` and `CONVERT` (6c).** Found by auditing the dictionary against the standard word lists. Core had no gaps (`."`, `S"` and `ABORT"` only looked missing because of how their names are spelled in the source). `CORE-EXT` was briefly reported complete at 6b while `S\"` and `CONVERT` were still missing; the audit caught it.
- ☑ **`SAVE-INPUT`** saves the source address, source length and `>IN` (three cells and the count 3). **`RESTORE-INPUT`** restores `>IN` only if the source address and length match, and otherwise returns true, so it cannot restore across `REFILL` or into a different `EVALUATE` string.
- ☑ **`S\"`** translates `\a \b \e \f \l \m \n \q \r \t \v \z \" \\` and `\xhh` into PAD, then shares the `S"` code path (label `SQSTATE`). More than 255 characters throws -18. **`CONVERT`** is `>NUMBER` with no length limit.
- ☑ **ANS test file `29_coreext_strings.tests.fs`.** A `T{ ... >R ... R> ... }T` line at top level reset the target (`>R` leaves the value under the interpreter's return address); the test now uses a definition.
- ☑ **MAME confirmation of stage 6c** (`S\"`, `CONVERT` and the section 29 tests): all unit tests and ANS tests pass. The section 29 file was then tidied by hand (formatting consistent with the other section files, base set to HEX).

ROM layout and unit tests

- ☑ **`BASECODE` and `BASEDICT` lowered** to `$DC44` and `$D395` (were `$DCC4` and `$D458`). The interrupt-driven build no longer fitted below `INITCODE` once `S\"` was added. The dictionary ends exactly at `BASECODE`.
- ☑ Polled `BASECODE` is 8659 bytes (`$DC44` - `$FE16`), interrupt-driven 8978 (`$DC44` - `$FF55`), `BASEDICT` 2223 (`$D395` - `$DC43`). Padding below `INITCODE`: 397 bytes polled, **78 bytes interrupt-driven**.
- ☐ **Interrupt-driven headroom is 78 bytes.** The next addition to the base code needs another downward shift of `BASECODE` / `BASEDICT` (the spare ROM below `BASEDICT` is about 5 KB, but the unit test code sits there too) or a saving.
- ☑ **Unit test group 11 split.** In the interrupt-driven build the StrParse tests overran `BASEDICT` by 17 bytes. `REPLACES`, `SUBSTITUTE`, `CLR-REPLACES`, `SNAME` and `UNESCAPE` now run as group 18 (`TSTSTRPARSE2`); group 11 keeps

the rest. Selector 0 calls both. `run all tests.sh` and `build all tests.sh` need `18` added to their selector lists.

Documentation

- ☑ `ClaudeForth.docx` and preview PDF dated 17 October 2026: 13 new glossary entries and index entries (`D0=` , `D0<` , `D2*` , `D2/` , `M*/` , `2LITERAL` , `2VALUE` , `.(` , `C"` , `S\"` , `CONVERT` , `SAVE-INPUT` , `RESTORE-INPUT`), updated `TO` and `ENVIRONMENT?` entries, the glossary count (242 words; the earlier text said 227 while 229 entries existed), memory map table and ROM sizes, selector table (18), test-suite text, Section 5 items and the regenerated source appendix (stage 6c). Memory-map SVG and PDF regenerated.
- ☑ README (and README_ANS.md): sections 27 to 29, query-table and double notes, `>R` guidance; the Double-Number entries removed from the open items.
- ☐ **Timeline** on the title page not updated.
- ☐ Older checklists (parts 3 and 4) still list the Double-Number gap as open; this part supersedes them.